THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM TWO JOINT EXAMINATION FOR GOVERNMENT SCHOOLS – JEGOS

PRE-MOCK EXAMINATION
032 CHEMISTRY
MARKING SCHEME 2024

SECTION A (15 Marks)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
B	D	B	A	C	D	D	A	C	B

(01@ = 10 marks)

2.

List A	(i)	(ii)	(iii)	(iv)	(v)
List B	F	D	B	C	H

(01@ = 05 marks)

SECTION B (70 Marks)

3. (a)

	The three states of matter		
Property	Solid	Liquid	Gas
Fluidity	Does not flow	Flow easily	Flow easily
Expansion on heating	Low	Medium	High ……0½ × 9 = 06 marks
Compressibility	Incompressible	Almost incompressible	Highly compressible
Motion of particles	Very Low	High	Very high

(b) Eeconomic importance of the changes in the states of matter.
(i) It enables separation of mixtures can through such processes as …
(ii) Industrial manufacture of products**:** Industrially, the process of distillation is applied in the production of pure substances such as …

(iii) Refining of petroleum (crude oil**):** Crude oil contains organic liquid components, each with a different boiling point. In the refinery, …

(iv) Drying of crops and clothes**:** When you suspend your clothing on a cloth line to dry, the moisture in it is lost through evaporation. Likewise, farmers …

(v) Cooling of our bodies in hot weather…,Ice formation in refrigerators…, Melting metals to make alloys…, formation of rainfall… ……01 × 4 = 04 marks

(Total: 10 marks)

4. (a)

(i) The components of air can be separated by physical means …

(ii) The components of are not in a fixed ratio

(iii) It does not have a chemical formula or specific chemical properties

(iv) Components are not chemically combined/formation not associated with heat changes from chemical reactions. 01½ @ = 06 marks

(b)

(i) To separate these substances, the solution is heated so that the solvent evaporates, leaving the solid residue behind. A mixture of salt in water can be separated by applying this method.

(ii) This is a method of mixture separation by evaporating off a liquid by means of distillation flask, and then condensing it by passing it down a water-cooled condenser and then collecting it as <u>distillate</u>. This method can be used to obtain pure water from impure water or from water with dissolved impurities. 02 @ = 04 marks

(Total: 10 marks)

5. (a)

(i) $0 = H + (Na^{+})$

$0 = H + 1$ ..... 00½ mark

$H = -1$ ..... 00½ mark

(ii) $0 = 2P + 5(O^{2-})$

$0 = 2P + (-10)$ ..... 00½ mark

$10 = 2P$

$P = +5$ ..... 00½ mark

(iii) $-2 = 2Cr + 7(O^{2-})$

$-2 = 2Cr + (-14)$ ..... 00½ mark

$+12 = 2Cr$

$Cr = +6$ ..... 00½ mark

(iv) $0 = Ba + 2(O^{2-})$

$0 = Ba + (-4)$ ..... 00½ mark

$Ba = +4$ ..... 00½ mark

(b) (i) Empirical formula

100% = %Carbon + %Hydrogen

% composition of carbon = 100% −20% = 80% ..... 00½ mark

Element	C	H
% composition	80	20
Atomic weight	12	1
Mole ratio	$\frac{80}{12} = 6.67$	$\frac{20}{1} = 20$
Relative mole ratio	$\frac{6.67}{6.67} = 1$	$\frac{20}{6.67} = 3$

..... 00½@ = 01 mark

..... 00½@ = 01 mark

The empirical formula is CH_3 ..... 00½ mark

(ii) Molecular formula

Molecular weight = Vapor density × 2 ..... 00½ mark

= 15 × 2 = 30

$30 = (CH_3)n$ ..... 00½ mark

$30 = (12 + (1 \times 3))n$

$30 = 15n$ ..... 00½ mark

n =2 ..... 00½ mark

$(CH_3)n = (CH_3)2 = C_2H_6$ ..... 00½ mark

The molcular formula is C_2H_6 ..... 00½ mark

(Total: 10 marks)

6. (a)
   i. availability
   ii. affordability.....@1 = 4 marks
   iii. high energy value/high heat content
   iv. easy to transport and store

(b) Data given .....01 mark

◦Initial temperature, θ1= 24.7℃ = 24.7 + 273 = 297.7 k

◦Final temperature, θ2= 68.5 ℃ = 68.5 + 273 =341.5 k

◦Mass of biodiesel, M = 56g = 0.056 kg

◦Volume of water, V =12 litres= 0.12 m3

◦Heat energy value of biodiesel, E = ?

Solution

◦From the formula,

◦But, Mass of water, m = density x volume .....01 mark

$=1000\ Kg/m^3 \times 0.012m^3 = 12$ kg

m =12 kg .....01 mark

$E = \boldsymbol{mc}\Delta$ .....01 mark

$E = 12x4.18x(341.5-297.7)0.056$

E = 392,32 KJ/Kg .....01 mark

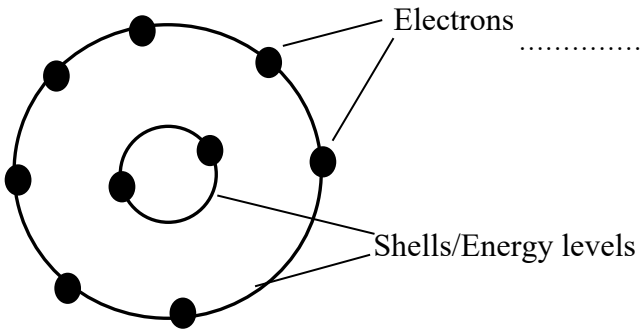
Heat energy value of biodiesel is 392,32 KJ/Kg. .....01 mark

(Total: 10 marks)

7. (a)
   (i). 9
   (ii) 12
   (iii) 21 .....01 @ = 04 marks

(iv) 2:7

(v) Energy-shells diagram of N



……0½ mark

……0½ mark

Diagram ……..02 marks

(b)

(i) Existence of atoms with the same atomic number but different mass……01 marks

(ii) Relative atomic mass = percentage of chlorine-35 + percentage of chlorine-37

$35 \times = \frac{75}{100} + 37 \times \frac{25}{100} = 35.5$ ……0½ mark

01 marks ……0½ mark

(Total: 10 marks)

8. (a) (i) Periodicity is the repetition of similar chemical properties of elements after a certain specific interval in a given arrangement. ……02 Marks

(ii) Periodic table is a tabular display of the chemical elements, which are arranged by atomic number, electron configuration, and recurring chemical properties .....02 Marks

(b)

(i) Alkali metals are Group 1 elements, they have one electron in the outer valence shell. Example; Lithium, sodium and potassium.

(ii) Alkaline earth metals are group 2 elements, they have two electrons in the outer valence shell. Example; Beryllium, magnesium and calcium.

(iii) The noble gases are group 8 elements. They have full outer shells. They have stable duplet (2 electrons) or octet (8 electrons) in their outermost shells which makes them very stable and therefore, unreactive. Examples are Helium and Neon .....02@ =06 Marks

(Total: 10 marks)

9.

i. Oxygen ……01 mark

ii. Potassium chlorate ⟶ Potassium chloride + Oxygen

$2KClO_{3(s)} \longrightarrow 2KCl_{(aq)} + 3O_{2(g)}$ ……02 marks

iii. Oxygen has almost the same density as air, so it cannot be collected by the upward displacement of air……02 marks

iv. Hydrogen peroxide and Manganese(IV)oxide……..01 mark

v.

- The oxygen in the air and that dissolved in water and soil is used for respiration.
- It is needed for burning.

- It is used in the oxyacetylene (oxygen–ethyne) flame for welding and cutting steel.
- It is extensively used for removing impurities from pig iron in order to produce steel. Oxygen is blown into molten iron to remove impurities such as carbon or phosphorus, which are expelled in the form of gases.
- Oxygen is used as an aid to breathing in hospitals, high altitude climbing or flying, and in deep sea diving.
- Liquid oxygen is used in supporting burning of fuels such as hydrogen in rocketsUsed in manufacture of hydrochloric acid …Any four 01@ = 04 marks

(Total 10 Marks)

SECTION C (15 MARKS)

10. Intro: Fires are classified based on the type of burning materials…..... 01½ marks

Class A fires: These are the fires in which the burning materials are ordinary combustible materials such as paper, wood, cardboard, coal, rubber, clothing, furniture and most plastics. Water is the best extinguisher for these fires. However, many other types of extinguisher may be used.

Class B fires: These fires involve flammable liquids such as petrol, kerosene, oil, alcohol, ether, vanishes, etc. For small fires, a fire blanket or sand may be used. If the fire is large, use foam, dry powder or carbon dioxide extinguisher. Water should not be used on class B fires because the burning material, being lighter than water, will just float and spread the fire further.

Class C fires: The burning material involves flammable gases such as hydrogen, acetylene, coal gas, butane, methane, and propane. The best fire extinguishers to use in fighting against these fires are foam, dry powder or carbon dioxide extinguishers. It is important to turn off the gas supply, and spray water on the gas tank to cool it down.

Class D fires: The burning material is a metal. Alkali metals such as sodium or potassium may catch fire when they come in contact with water and oxygen. At high temperatures, many metals react with oxygen vigorously. Fires that involve burning metals should not be extinguished by water. This is because the burning metal can react with water to give hydrogen (another potential fuel). The appropriate extinguisher to use is foam or dry powder extinguisher.

$6 \times 02@ = 12$ marks

Class E fires: These fires involve electrical equipment such as appliances, wiring, circuit breakers and outlets. You may use carbon dioxide or dry powder extinguisher to put off these fires. Never use water as it can conduct electricity and give an electric shock.

Class F fires: The burning material is cooking oil or fat. A cooking oil fire in the kitchen can be extinguished by covering the pan with a fire blanket or damp cloth. Foam, dry powder or carbon dioxide extinguishers also work by cutting off the air supply to the fire. For large fires, wet chemical extinguishers are recommended.

Conclusion: Given the basis for classiying the fires, therefore it is important to know the size and class of the fire that you are going to put off before starting to fight it .....1½ marks